

Manuscript

Layer-1

稿件草稿 HTML 版（基于 docs/manuscript_draft.md）

Self-contained academic HTML with embedded SciencePlots figures (base64).

Literature-anchored Layer-1 surrogate for exploratory screening of indirect mitral annuloplasty

Working short title: FMR IMA Layer-1 exploratory surrogate (software / methods)

Venue class: computational biomechanics / research-software methods paper (CMBBE / MedEngPhys / RSOS-class) — not a clinical decision trial.

One-sentence argument: We provide a transparent, literature-anchored algebraic surrogate that reproduces selected Galili peak-systole anchors, supports Dryad-aware provenance, reports true fold-wise response-model cross-validation (and a separate blend-off diagnostic), and ranks exploratory IMA-CS / IMA-AP scenarios under explicit geometric and η assumption-prior uncertainty — without claiming new LHHM FEA, clinical recommendation, or patient-level external validation.

Data / reproduce: `python run_pipeline.py --seed 42 --paper · python tools/import_galili_dryad.py --from-fixture --process · python -m analysis.fit_response_model --write · python tools/loo_evaluate.py --write`

Verification levels: Level 0 = calibrated to / reproduced selected Galili cases; Level 1 = exploratory planning map + uncertainty-aware scenario ranker + fold-wise response diagnostic. Level 2 patient-specific LHHM/FSI is out of scope.

Abstract

Background. Computational IMA studies (Galili et al., *R. Soc. Open Sci.* 2022; doi:10.1098/rsos.211464; title: *Numerical biomechanics modelling of indirect mitral annuloplasty treatments for functional mitral regurgitation*) report suture/bridge shortening with peak-systolic geometry and leakage. Peak-systolic IMA-AP 50% yields AP **15.9 mm** (near-direct AP effect), not undeformed diastole **34.4 mm**. Device classes differ: MAVERIC evaluates **ARTO** (CS-IAS / IMA-AP-class); Carillon / REDUCE-FMR are IMA-CS-class.

Methods. We implement a reproducible Python Layer-1 surrogate: phenomenological mechanics plus a literature-calibrated leakage proxy, with a config-switched response path (`fitted_response` default for paper ranking; `rule_based_proxy` retained for diagnostics). Dryad (doi:10.5061/dryad.bzkh1899d) supplies primary provenance **and** an auxiliary model feature `contact_fraction` (interpolated for off-table design points); scalar AP/ROA/leakage remain table-backed unless derivable. We fit fold-wise response models `f_ROA( $\Delta$  AP, device family, annular reduction, contact_fraction)` and `f_leak(ROA, coaptation proxy,`

contact, device mechanism) under true leave-one-case-out, and a full-train fit for ranking. A discrete scenario ranker minimizes leakage_proxy_pct subject to an AP ceiling, an illustrative NiTi engineering screen, and a Rottländer CS–LCx risk-screening threshold, reporting device-candidate P(feasible), Latin-Hypercube multi-parameter ranking stability (η_{AP} , η_{CS} , dual factor, CS–LCx baseline/slope; $N \approx 120$), and family-separated Pareto frontiers. Dual-suture commissural factor is an exploratory hypothesis **sensitivity** parameter (Fig. 5: 0.25/0.5/0.75/1.0), not a discovery claim. Ranking is assumption-driven exploratory analysis, not validated prediction.

Results (seed=42). Fold-wise response-model CV on the held-out Galili subset improves ROA MAE from rule-based blend-off **50.18 mm²** to about **6.37 mm²** and leak MAE from **1.445** to about **0.245** percentage points; IMA-AP 70% absolute ROA/leak errors fall from **94.17 mm² / 5.455 pp** to about **6.60 mm² / 0.198 pp** (internal engineering targets, not medical validation). Further dampening of stacked rule-based AP70 penalties reduces Galili-mapped AP70 leak from historical $\approx 2.46\%$ to $\approx 1.21\%$ (still overestimates Galili **0.13%**; paper ranking uses fitted_response). Under the planning map the ranker evaluates **36** total points (**35** device candidates; pathology is not a candidate) and retains **30** feasible designs (**p_feasible_device_candidates** ≈ 0.857). With fitted_response, seed-42 best candidate is **IMA-CS 20%** (AP↓ 11%, leakage proxy $\approx 0.39\%$); dual-suture AP60 remains an assumption-sensitive alternative under the nominal dual hypothesis (not Innovation D discovery). LHS UQ ($N=120$) yields **P(top-1)** ≈ 0.34 and mean feasible fraction ≈ 0.82 .

Conclusions. A literature-anchored low-order surrogate can support **exploratory screening** of IMA strategy settings with inspectable assumptions, honest held-out reporting, and uncertainty language. It is not a preoperative clinical decision system and must not equate physics % with clinical regurgitant volume.

Keywords: functional mitral regurgitation; indirect mitral annuloplasty; surrogate model; exploratory screening; research software; Layer-1

1. Introduction

Functional mitral regurgitation (FMR) reflects ventricular remodeling, annular dilatation, and impaired leaflet coaptation. Indirect mitral annuloplasty (IMA) concepts remodel annular geometry without direct leaflet repair. **IMA-CS** (Carillon-class) and **IMA-AP** (CS–IAS; ARTO / MAVERIC mechanism class) are not mechanically interchangeable.

Galili et al. (2022) compared generic IMA-CS and IMA-AP in the Living Heart Human Model. IMA-AP acts nearly directly on peak-systolic AP (disease 26.1 mm → IMA-AP 50% → **15.9 mm**). Undeformed diastolic AP 34.4 mm is a different cardiac phase.

What remains for software methods is transparent **dose translation, provenance, held-out honesty, and exploratory ranking** under stated assumptions—not a claim of first CS-vs-AP comparison, not new FEA, and not a clinical recommendation engine.

2. Related work and positioning

Galili RSOS 2022 (doi:10.1098/rsos.211464) + Dryad supporting files provide the computational leakage/ROA/AP table we anchor to. Clinical series (ARTO/MAVERIC; Carillon TITAN II; REDUCE-FMR) supply **directional / geometric context only**. This paper contributes research-software packaging, fold-wise response diagnostics, and an inspectable Layer-1 ranker, not a new high-fidelity FSI study.

3. Software architecture

```
configs/    surrogate_calibration.yaml, design_space.yaml, case YAML
models/     geometry, pathology, devices, response_model (fold-wise f_ROA / f_leak)
simulation/ phenomenological mechanics proxy (legacy alias run_fea_surrogate)
sph/        literature-calibrated leakage proxy (SPH-inspired; not a full SPH solver)
analysis/   ROA, jet, sweep, scenario_ranker, fit_response_model, paper tables/plots
tools/      import_galili_dryad.py, loo_evaluate.py (blend-off diagnostic), package_reports.py
data/       raw drop zone, fixtures, processed galili_cases.csv
results/output/cross_validation/  true fold-wise CV artifacts
```

```
Runnable entry points: run_pipeline.py, python -m analysis.planner, python -m
analysis.fit_response_model, python -m analysis.design_sweep.
```

4. Methods

Pathology and geometry. Parametric papillary pathology (44% posterior passive). Undeformed diastole and peak systole are separate cardiac_phase records. AP diameter is a literature-defined geometry **input** (ap_role: literature_mapping_input); AP MAE is not reported as predictive accuracy.

Device kinematics. Galili mode interpolates published peak-systolic AP; clinical/planning mode uses assumption-prior η or preferred ΔAP . η_{CS} is **not** fitted from MAVERIC/ARTO. Default uncertainty uses Latin Hypercube over $\eta_{AP} \in U(0.24, 0.36)$, $\eta_{CS} \in U(0.44, 0.66)$, dual commissural factor, CS-LCx baseline, and cinch slope (N documented in design_space.yaml).

Mechanics and leakage proxies. Algebraic coaptation-gap / strain / contact proxy scores. Leakage proxy is literature-calibrated at pathology; `_sph_scale` is the pathology regurgitation fraction identity. Physics regurg % $\neq$ clinical regurgitant volume. Synthetic contact-cluster ROA is visualization-only (model_weight=1, cluster_weight=0); Dryad contact_fraction enters the fitted response path as a science feature. Official export fields: leakage_proxy_pct, strain_risk_score, contact_score (legacy aliases retained).

Response paths. configs/design_space.yaml response_path: fitted_response (paper/seed-42 default), rule_based_proxy (legacy exploratory), or hybrid_ap_extreme (rule-based with AP-class over-short routed to fitted).

Provenance. `tools/import_galili_dryad.py` downloads or documents manual drop; extracts Dryad `contact_fraction` and `n_sph_particles`. Scalar AP/ROA/leakage remain table-backed (not invented from unlabeled leaflet nodes).

Diagnostics vs true CV. `tools/loo_evaluate.py` is an **anchor-free / leave-one-case blend-off diagnostic** on the fixed rule-based surrogate (not fold-wise refit). `analysis/fit_response_model.py` performs **true fold-wise** fitting with held ROA/leak excluded from train arrays. Full-train params for ranking live under `results/output/response_model/`.

Scenario ranker. Minimizes `leakage_proxy_pct`; reports `n_total_points`, `n_device_candidates`, `n_feasible_device_candidates`, `p_feasible_device_candidates`, `P(top-1)`, and first-order Spearman sensitivity ranks. Pareto families: `pareto_global_common_objectives` (min leak, max AP↓), `pareto_ima_cs_family` (adds LCx/NiTi), `pareto_ima_ap_family`. Missing LCx/NiTi are never filled with 1e9/0. Dual-suture factor is an explicit hypothesis **sensitivity** (Fig. 5), not discovery. LCx prefers patient-measured baseline. The 14–20% AP band is a clinically contextualized **exploratory planning range** (anchor ~14–15%, ceiling 20%; legacy `clinical_window` alias only).

5. Results

5.1 Data provenance

Dryad archives yield deformed-contact and SPH coordinate files. Processed table `data/processed/galili_cases.csv` marks `source_tier=dryad_derived` for extracted `contact_fraction` and `n_sph_particles`. Published peak-systole AP, ROA, and leakage scalars remain table-backed fills — the importer does **not** extract independent AP, ROA, and leakage descriptors from unlabeled coordinates alone.

5.2 Calibration reproduction

Case	Leakage %	Peak-sys AP mm	ROA mm ²
pathology	5.26	26.1	172.8
ima_cs_22	0.29	24.8	48.2
ima_ap_50	0.08	15.9	27.3
ima_ap_70	0.13	12.4	46.1

Calibration IDs with blend ON are reproduction only.

5.3 Anchor-free diagnostic and true fold-wise CV (includes AP70 failure/improvement)

Rule-based blend-off diagnostic (`tools/loo_evaluate.py`, held-out IDs): historical pre-dampen baseline MAE ROA **50.18 mm²**, leak **1.445 pp**; IMA-AP 70% previously **140.3 / 5.585%** (abs err **94.17 / 5.455**).

Current dampened rule-based held-out MAE $\approx 28.6 \text{ mm}^2 / 0.66 \text{ pp}$. Critically, the rule-based path **captures a qualitative non-monotonic tendency** (Galili AP50 leak 0.08% $\rightarrow$ AP70 0.13%) but **substantially overestimates AP70 leak magnitude** (pred $\approx 2.46\%$ vs published 0.13%; abs err $\approx 2.33 \text{ pp}$) — this is **not** an unqualified claim that the surrogate “reproduces Galili non-monotonic behavior.”

True fold-wise response-model LOO (analysis/fit_response_model.py $\rightarrow$ results/output/cross_validation/): held-out-subset MAE ROA $\approx 6.4 \text{ mm}^2$, leak $\approx 0.25 \text{ pp}$; AP70 abs err $\approx 6.6 \text{ mm}^2 / 0.20 \text{ pp}$. Internal engineering targets (ROA MAE < 25 , leak MAE < 0.5 , AP70 ROA < 25 , AP70 leak < 0.5) are met on this seven-case table — **not** patient-level external validation. Full seven-fold MAE remains inflated by the pathology leave-out (large ROA scale gap). AP is literature geometry input (ap_prediction_metric_applicable=false).

Full held-out table (CS14 / CS18 / AP30 / AP70) — published vs rule-based blend-off vs fold-wise response model:

Case	Pub ROA	Rule ROA (abs err)	Fold ROA (abs err)	Pub leak %	Rule leak (abs err pp)	Fold leak (abs err pp)
ima_cs_14	56.7	24.0 (32.7)	69.6 (12.9)	0.52	0.65 (0.13)	0.60 (0.08)
ima_cs_18	55.3	17.6 (37.7)	53.4 (1.9)	0.41	0.34 (0.07)	0.42 (0.01)
ima_ap_30	51.1	14.9 (36.2)	47.0 (4.1)	0.16	0.28 (0.12)	0.86 (0.70)
ima_ap_70	46.1	~ 35.6 (~ 10.5)	39.5 (6.6)	0.13	~ 1.21 (~ 1.08)	0.33 (0.20)

AP70 row is reported front-and-center: fold-wise meets internal engineering screens; dampened rule-based leak improved from historical $\approx 2.46\%$ but still overestimates Galili 0.13% — paper ranking prefers fitted_response.

5.4 Exploratory scenario ranking (planning map, seed=42; fitted_response)

- **n_total_points=36, n_device_candidates=35, n_feasible_device_candidates=30, p_feasible_device_candidates ≈ 0.857**
- With default response_path=fitted_response, best candidate ranked **IMA-CS bridge 20%** ($\eta_{\text{cs}}=0.55 \rightarrow$ AP reduction **11.0%**, leakage proxy $\approx 0.39\%$, jet=central, CS-LCx **8.6 mm** at risk-screen edge)
- Dual-suture AP60 remains an assumption-sensitive alternative under the **nominal dual-suture hypothesis** (Fig. 5 = factor sensitivity 0.25/0.5/0.75/1.0, **not** Innovation D discovery)
- LHS multi-parameter UQ (N=120): **P(top-1) ≈ 0.34** , mean feasible fraction ≈ 0.82 ; first-order Spearman ranks dominated by CS-LCx baseline/slope
- Pareto: global common objectives; CS/AP family frontiers without N/A fillers

5.5 Directionality-only clinical context

ARTO/MAVERIC: AP↓. Carillon TITAN II: ~15% AP context. REDUCE-FMR: regurg↓. Magnitudes not equated to physics %.

6. Discussion / limitations

Central advance: transparent exploratory screening software with corrected Galili peak-systole AP semantics, MAVERIC=ARTO mapping, Dryad `contact_fraction` in the fitted path, honest held-out/CV reporting (including AP70), LHS ranking stability, and family-correct Pareto language.

Irreducible limits (documented, not faked):

- **n=7** Galili peak-systole table cases — not patient-level external validation
- **No patient CT** / patient-specific chamber geometry
- **No chamber-labeled SPH leakage recompute** from Dryad particle clouds (full-domain cloud only; AP/ROA/leak remain table-backed)
- Algebraic / phenomenological proxies; η and dual factor are assumption priors
- LCx/NiTi screens are literature/engineering screens, not safety clearances
- Rule-based AP70 still overestimates leak magnitude vs Galili even after dampening; prefer fitted path for ranking

7. Availability

- Code: <https://github.com/Coucou2016/fmr-ima-layer1-planner>
- License: MIT (LICENSE)
- Citation: CITATION.cff · references.bib
- Model card: MODEL_CARD.md · Data: DATA_PROVENANCE.md

8. Conclusion

A Layer-1 literature-anchored surrogate can rank exploratory IMA scenarios under explicit assumptions after correcting peak-systole Galili AP facts and MAVERIC device-class attribution, and after reporting true fold-wise response diagnostics alongside a renamed blend-off casewise diagnostic. Cite as methods/software for exploratory screening—not as clinical preoperative decision support.

Collab *notes:* docs/chatgpt_collab/20260913_p0_review_response.md,
docs/chatgpt_collab/20260914_plp2_implementation.md,
docs/chatgpt_collab/20260914_round2_major_revision.md,
docs/chatgpt_collab/20260915_r2_major_revision.md

Embedded SciencePlots figures

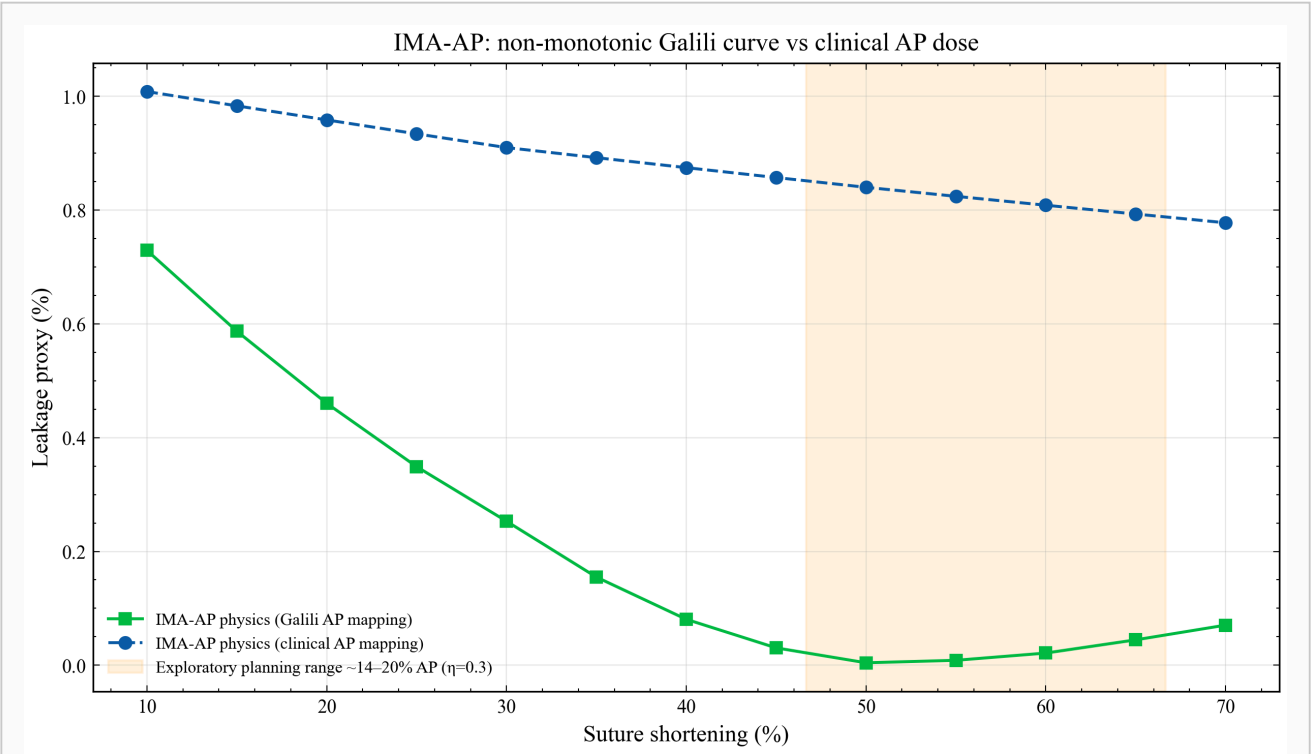


图 1. IMA-AP 物理反流随缝线缩短百分比的变化：Galili 映射 vs 临床映射

文件: fig1_ima_ap_nonmonotonic_exploratory_planning_range.png (SciencePlots + Times New Roman, dpi≥300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉与读图说明：本图回答“计算缝线缩短百分比能否直接当作临床前后径（AP）剂量？”横轴为 IMA-AP 缝线缩短 %；纵轴为 Layer-1 泄漏代理 leakage_proxy_pct（代数力学代理或 fitted_response；不是超声反流容积分数）。

如何读：（1）Galili 映射复现发表峰缩期 AP：IMA-AP 50% → 峰缩期 AP=15.9 mm（相对疾病 26.1 mm 约 39% AP 缩减），并非未变形舒张期 34.4 mm / 0% AP；70% → 12.4 mm 并交界区泄漏升高。（2）规划映射采用假设先验 $AP_reduction\% = \eta \times shortening\%$ ，默认 $\eta_{ap}=0.30$ （假设，非临床标定），使 50% 缝线对应约 15% AP，落入 ARTO/MAVERIC（IMA-AP 类，不是 Carillon）约 14–15% 对话窗口。绿色带为 exploratory planning range 约 14–20% AP。

结论（仅限本代理）：规划映射下窗口内总体可筛查；Galili 峰缩期映射下 70% 非单调恶化。勿将纵轴等同临床试验反流容积。

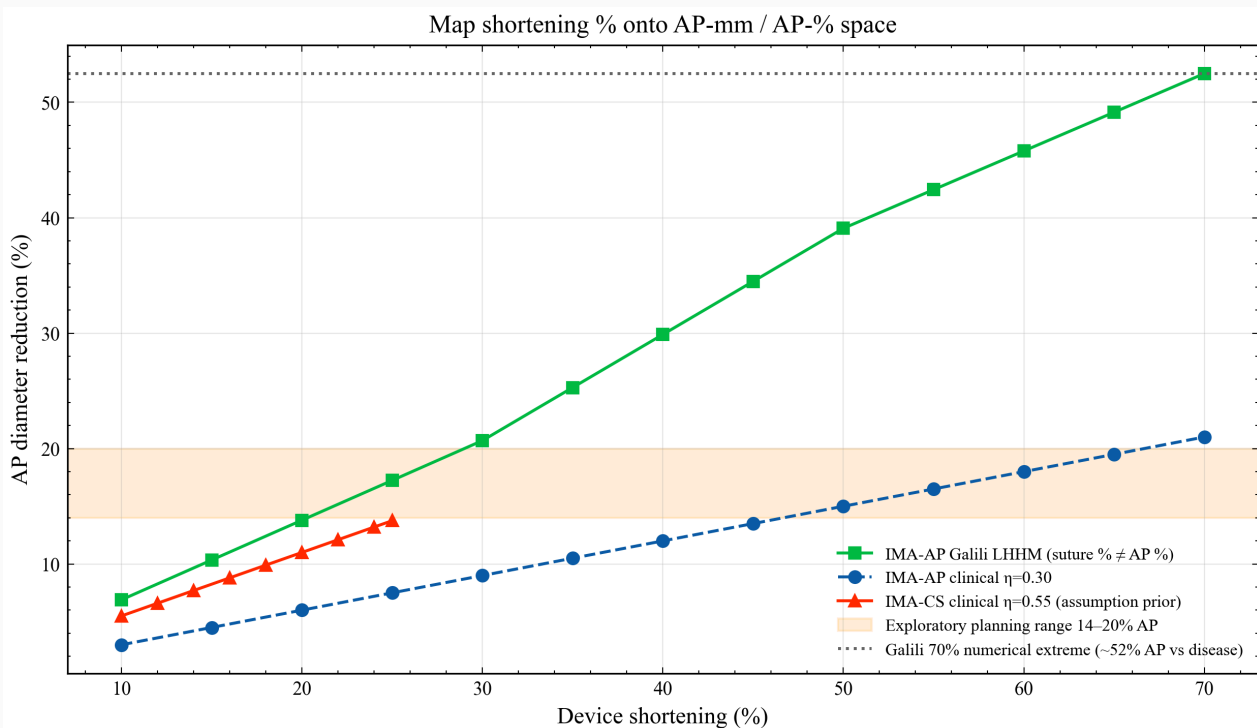


图 2. 装置缩短百分比到 AP 直径缩减百分比的剂量映射

文件: fig2_suture_vs_ap_reduction.png (SciencePlots + Times New Roman, dpi $\geq$ 300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉与读图说明: 横轴为缩短 %; 纵轴为 AP 缩减 %。轨迹对照: Galili 峰缩期 (IMA-AP 近直接缩 AP)、

规划 IMA-AP (假设 $\eta_{ap}=0.30$)、规划 IMA-CS (假设 $\eta_{cs}=0.55$, 非 MAVERIC 拟合; Carillon/TITAN II $\sim 15\%$ 仅语境)。

MAVERIC=ARTO (IMA-AP 类)。水平带 14–20% 为探索性规划窗口。

如何读: 同一缩短 % 在不同映射下 AP 缩减截然不同。 η 为假设先验, 不是成像–力学辨识, 也不是“临床标定常数”。

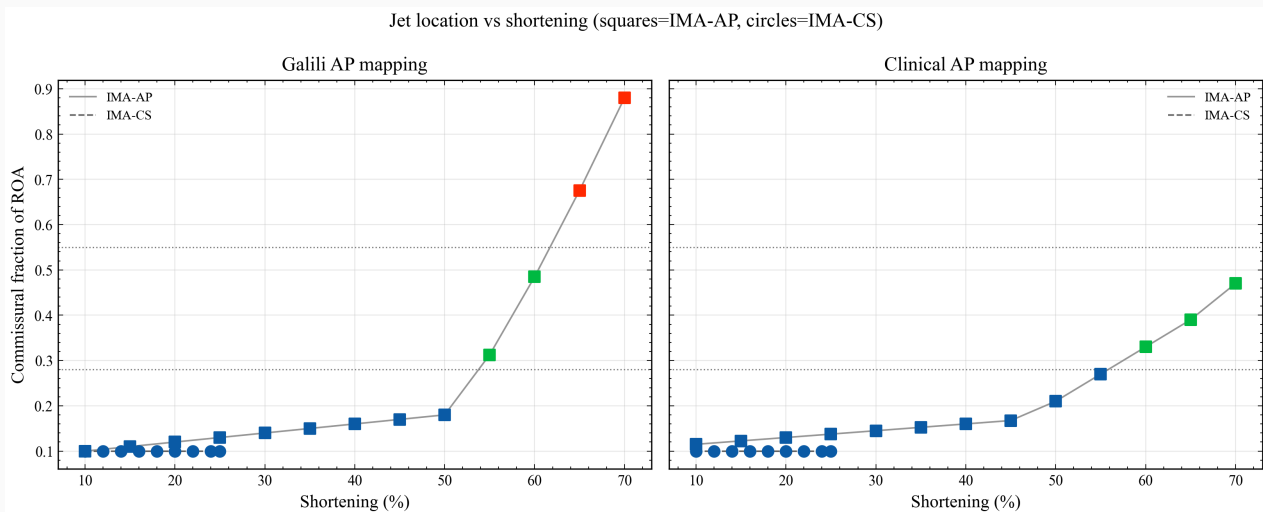


图 3. 射流位置与交界区 ROA 分数随缩短变化（Galili vs 临床）

文件: fig3_jet_location.png（SciencePlots + Times New Roman, dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉与读图说明：反流“量”之外，筛查还关心泄漏机制位置。本代理将 ROA（regurgitant orifice area，

反流口面积，单位 mm²，由对合间隙等代理量估计）拆为中央份额与交界区份额，并分类为 central / commissural / mixed。横轴为缩短 %；纵轴为交界区 ROA 分数；方块=IMA-AP，圆点=IMA-CS；左右面板对照

Galili 与临床映射。水平虚线为分类阈值示意。双缝线 ×0.5 为显式假设参数，不是独立“发现”。

如何读：交界区分数升高意味着泄漏更偏交界区；分类从 central 变为 mixed/commissural 表示机制标签切换。

该标签是代理口上的机制草图，不是超声 PISA（proximal isovelocity surface area）或多普勒诊断。

seed-42 假设下最优候选双缝线 60% 的机制来源可在本图与图 5 对照中追溯：同 AP 下交界份额更低、jet 保持 central。

结论：在临床映射下，IMA-AP 高剂量单缝线更易出现 mixed；IMA-CS 在可行桥缩短范围内多保持 central。

结合图 5，双缝线可在匹配 AP 缩减下压低交界区分数（假设情景）。

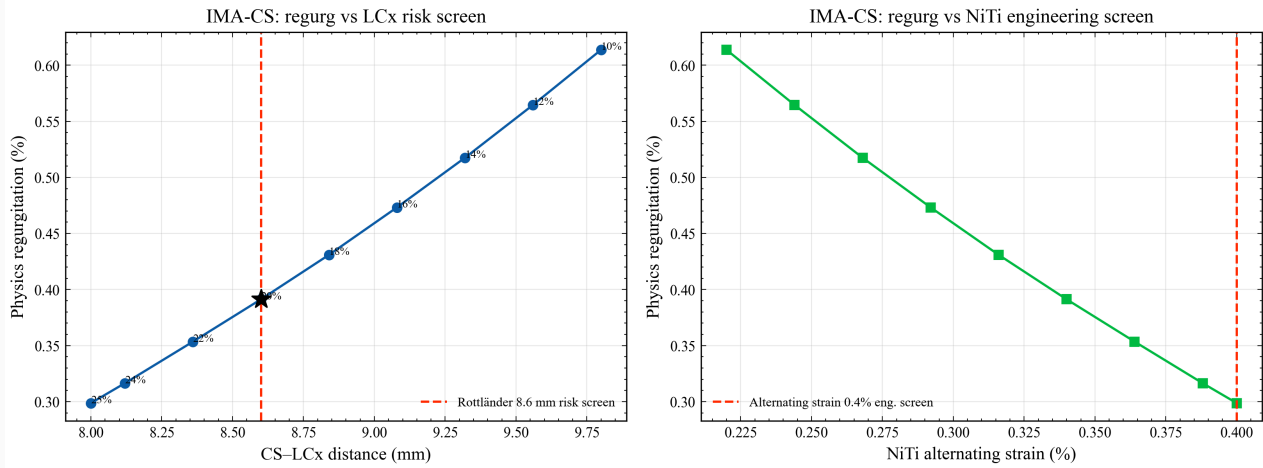


图 4. IMA-CS 临床映射：反流–筛查 Pareto（CS–LCx 与 NiTi 交变应变）

文件：fig4_pareto_lcx_strain.png（SciencePlots + Times New Roman，dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉与读图说明：IMA-CS 在降低反流的同时可能压缩 CS–LCx 距离（coronary sinus–left circumflex artery，

冠状窦与左回旋支间距）。文献筛查边界取远端着陆区 CS–LCx ≥ 8.6 mm（Rottländer 等；注意：<8.6 mm

预测妥协 ≠ ≥8.6 mm 已证明安全）。同时对 NiTi（镍钛合金桥）交变应变设置 <0.4% 的工程筛查上限。默认示意解剖基线 CS–LCx=11.0 mm，可替换为患者 CT。标题用“筛查”而非“安全证明”。

如何读左右面板：横轴多为筛查指标（CS–LCx 或交变应变），纵轴为 physics 反流；点随桥缩短移动，越接近约束边界，可行域越窄。22% 及以上在默认解剖上常因 CS–LCx 违约束而不可行。

结论：默认解剖上可行最优贴近 CS–LCx=8.6 mm 的桥缩短 20%。该结果是规划器筛查信号，不是患者级安全证明。

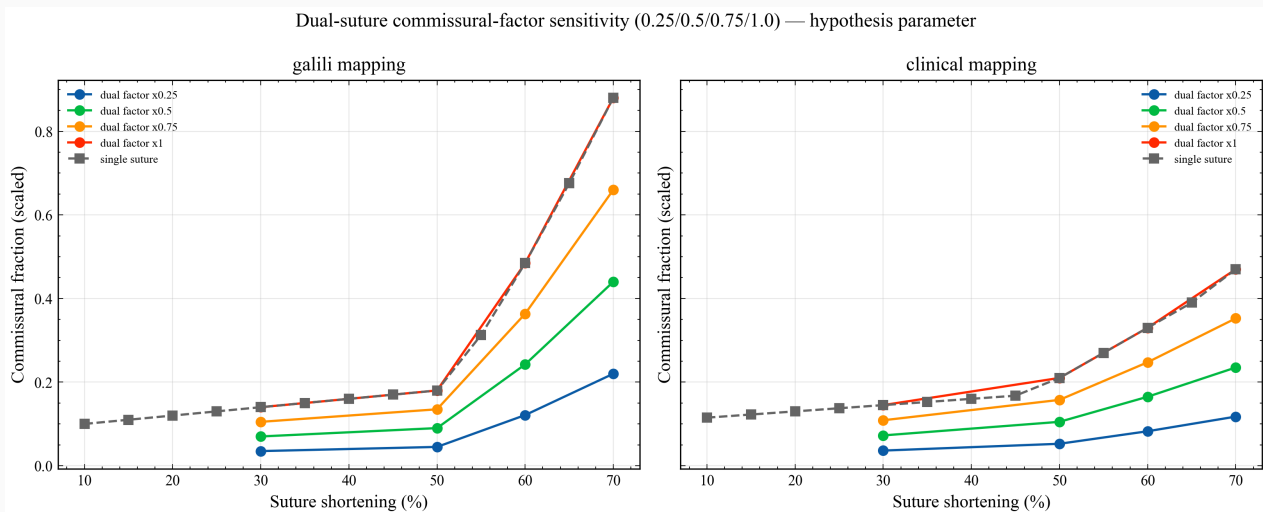


图 5. 双缝线交界因子敏感性（0.25/0.5/0.75/1.0）——假设参数，非 **Innovation D** 发现

文件: fig5_dual_vs_single_suture.png (SciencePlots + Times New Roman, dpi≥300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉与读图说明: 本图展示双缝线交界因子 (0.25/0.5/0.75/1.0) 敏感性, 不是“**Innovation D discovery**”。

名义因子 $\times 0.5$ 为显式假设参数; 排序对因子选择敏感。勿将 physics 泄漏差 (如 dual60 vs single ~ 0.0018 pp) 写成临床反流容积改善。